



SUMMER INTERNSHIP PRESENTATION

RESEARCH INTERNSHIP AT NIT TRICHY

By

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In fulfilment of Summer internship for the award of the degree

Bachelor of Technology

in

Electronics and Communication Engineerin

About the Internship:

- Institute: NIT Trichy
- Duration: 12th May – 10th July 2025
- Mentor: Dr. Varun Gopi
- Guide: Mr. Bibin
- Objective: ECG classification using Deep Learning

BIOLOGICALLY INSPIRED NEURAL NETWORK FOR CARDIOVASCULAR DIAGNOSIS

INTRODUCTION:

- Cardiovascular diseases are the leading cause of death globally.
- CardioECGNet mimics the human heart's electrical conduction system.
- Processes ECG images to classify various cardiovascular conditions.
- Designed to improve interpretability and performance in biomedical AI.
- Enable reliable binary classification for various heart-related conditions.
- Uses layers inspired by SA node, AV node, Bundle of His, and Purkinje fibers.

Motivation / Problem Statement:

Challenges:

- ECGs often contain **noise, artifacts, baseline wander**
- Raw 1D classification → limited by **loss of time-frequency info**
- **Class imbalance** across normal vs abnormal beats

Our Approach:

- **Raw 1D ECG Classification**
 - Direct waveform input to 1D CNN
 - Captures temporal patterns but noise-sensitive
- **2D Scalogram Classification**
 - Convert ECG → **time-frequency scalograms** (CWT)
 - 2D CNNs exploit spectral + spatial features
 - More robust, better accuracy

Objectives:

- Automate ECG arrhythmia detection
- Improve accuracy and robustness
- Enhance interpretability of models

Datasets Used:

Dataset	Leads	Size	Description
MIT-BIH Arrhythmia Database (Massachusetts Institute of Technology-Beth Israel Hospital)	5-lead	48 half-hour recordings (~109,000 beats)	Benchmark dataset for arrhythmia detection, includes annotated normal and abnormal heartbeats
PTB-XL Database (Physikalisch-Technische Bundesanstalt - eXtended Large)	12-lead	21,837 10-sec clinical ECG records	Large-scale real-world dataset with multiple diagnostic labels (normal & abnormal), supports deep learning studies

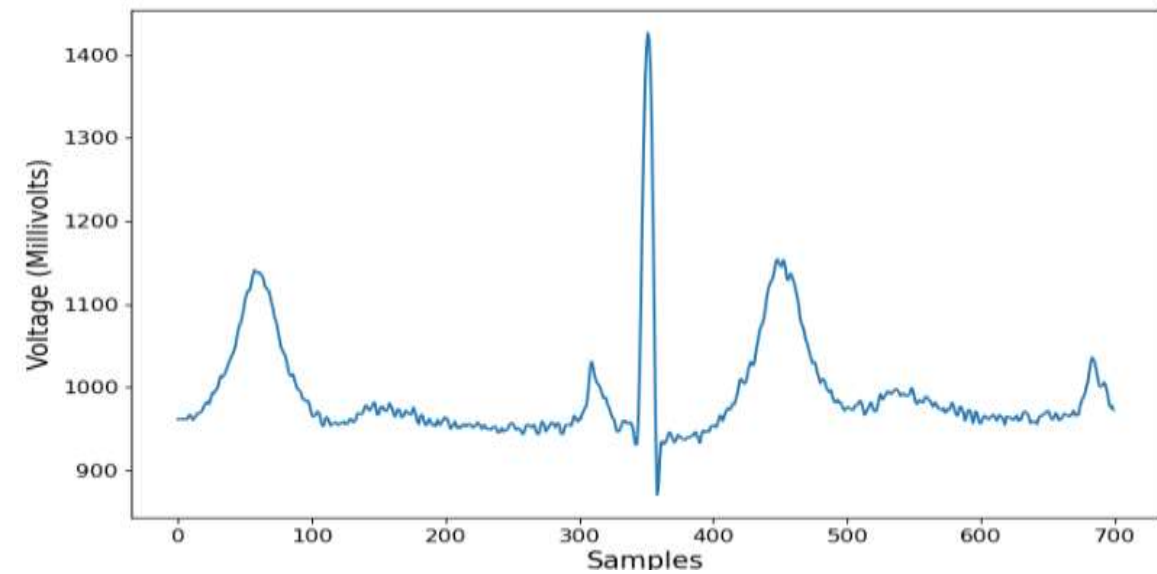
MIT-BIH Arrhythmia Database

Methodology:

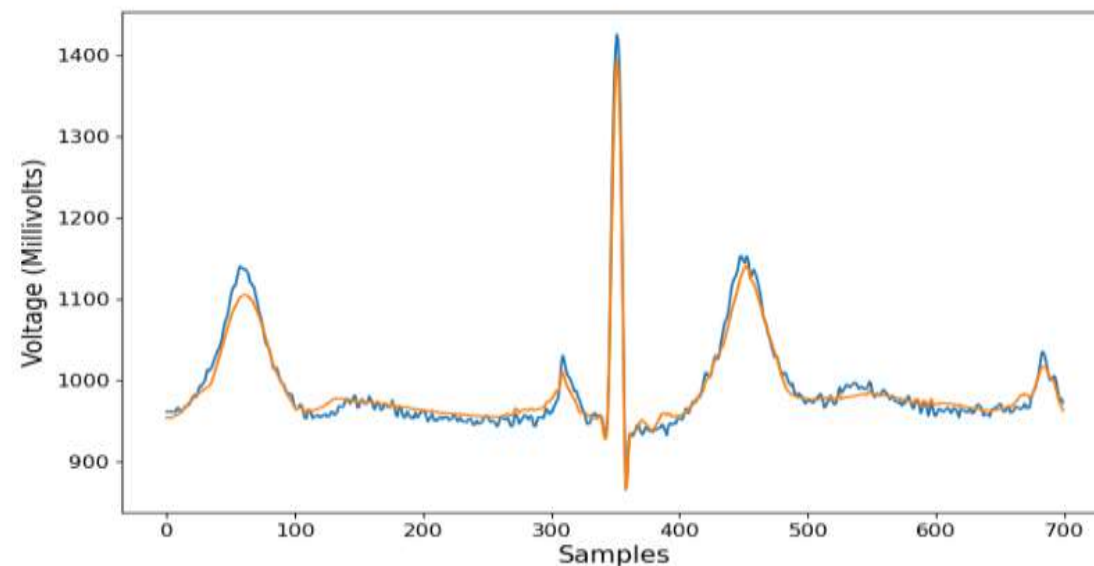
Data Preprocessing

- **Noise issues:** baseline wander (respiration), power-line interference (50/60 Hz), muscle artifacts
- **Steps applied:**
 - Bandpass filter → removes drift & high-frequency noise while keeping QRS.
 - Wavelet denoising (DWT) → suppresses noise, preserves morphology.
 - R-peak detection , segmentation into fixed-length beat windows.
- **Normalization** : ensures standardized input for deeplearning.
- **Data Balancing:**
 - Original dataset highly imbalanced (Normal >> Arrhythmic beats)
 - Applied **random under-sampling of majority class (N) + over-sampling of minority classes**
 - Ensured **balanced training set** for fair model learning

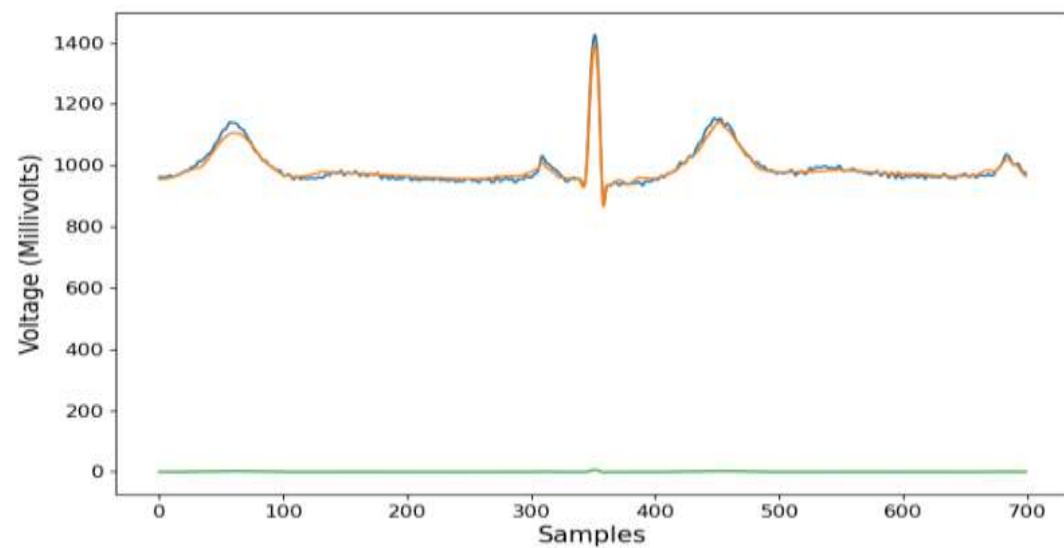
Before preprocessing:



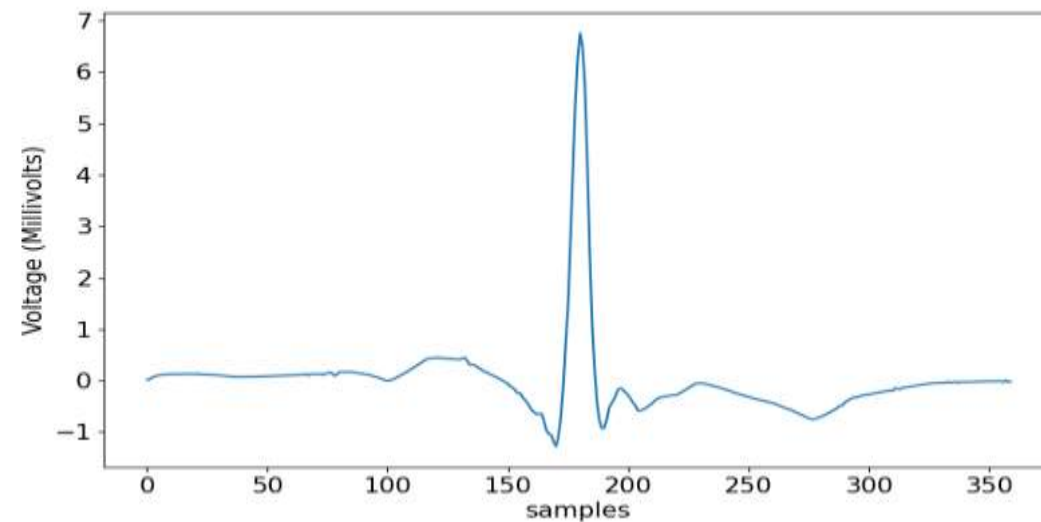
After Denoising:



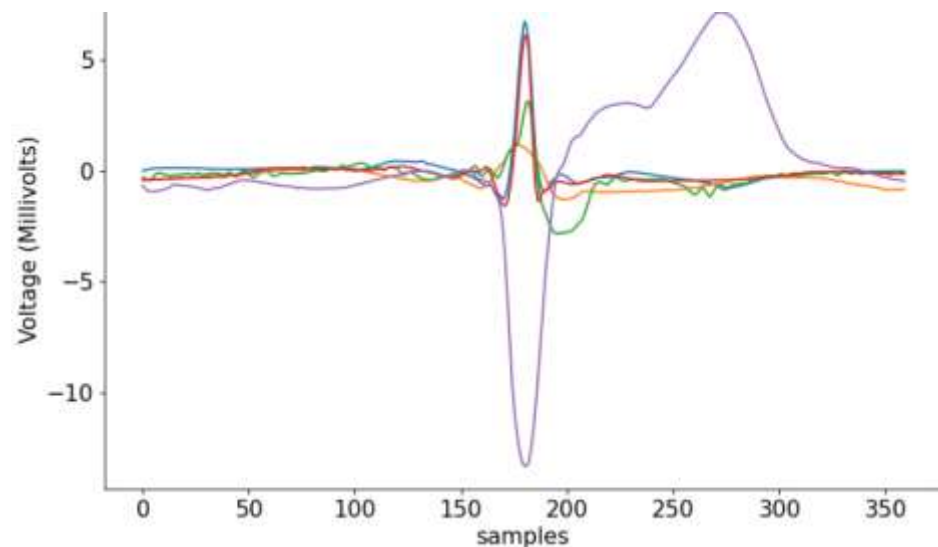
After Normalization:



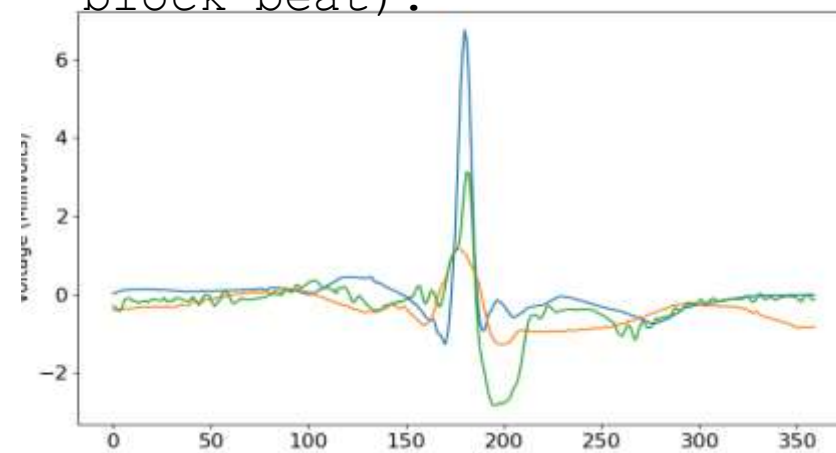
N_Normal beat



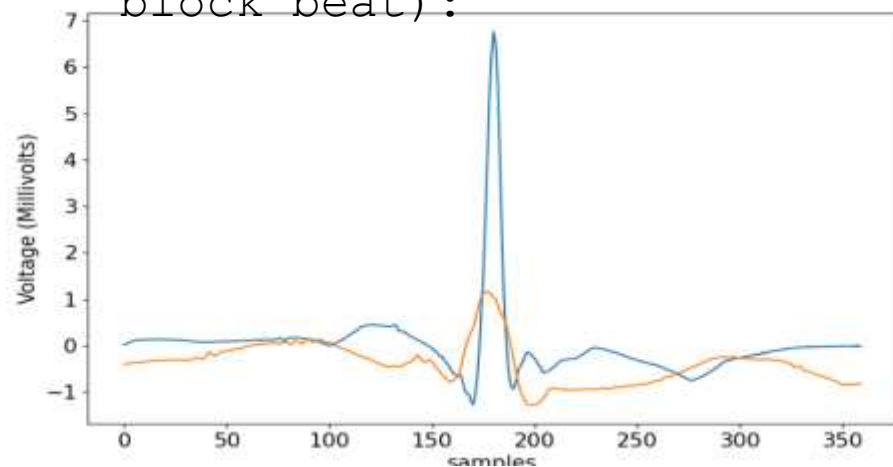
V (Premature ventricular contraction beat):



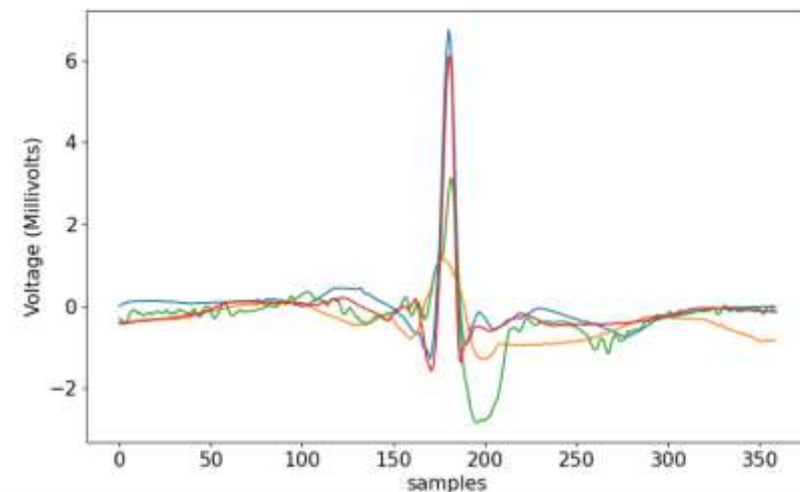
R (Right bundle branch block beat):



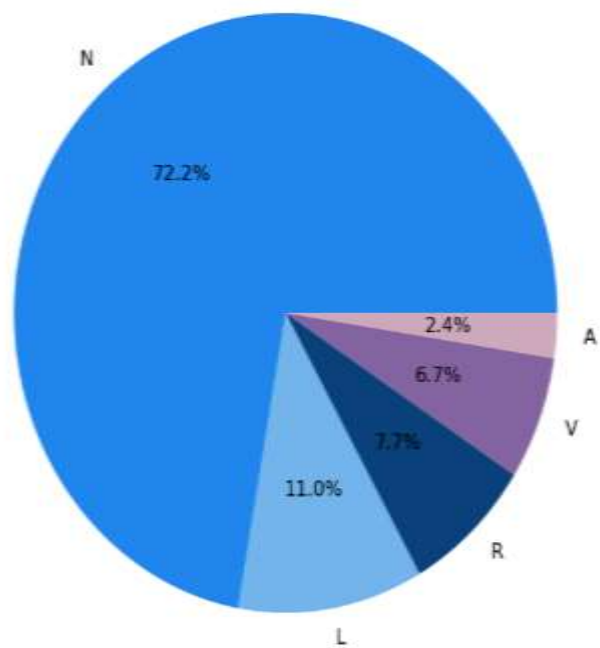
L (Left bundle branch block beat):



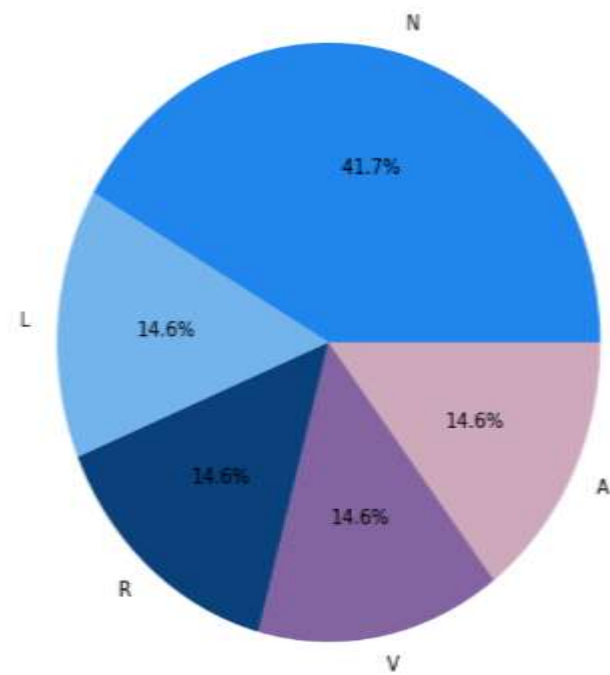
A (Atrial premature beat):



Before Balanced:



After Balanced:



Feature Representation:

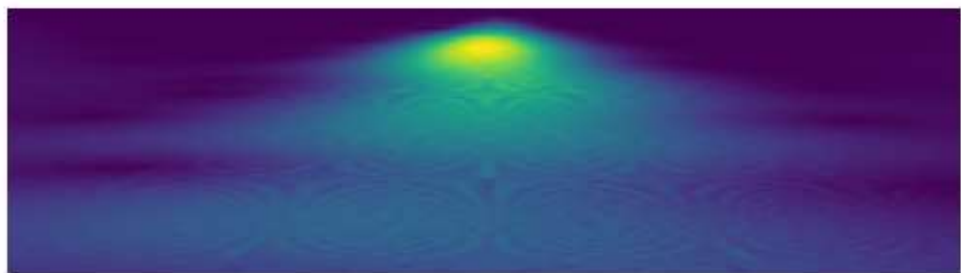
1D Signal Representation:

- Direct raw beat signals (time-domain)
- Input to **1D CNN** → captures **temporal amplitude patterns**
- Pros: Preserves original waveform
- Limitation: Sensitive to noise, poor class separation for similar morphologies

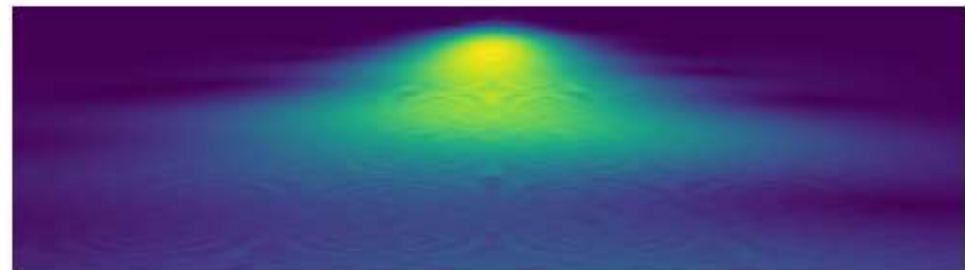
2D Scalogram Representation:

- Beats converted to **scalograms** using Continuous Wavelet Transform (CWT)
- Provides **time-frequency view** of ECG
- Captures subtle changes in morphology & timing
- Input to **2D CNN** → extracts spatial & spectral features like images
- Advantage: Improves discrimination between

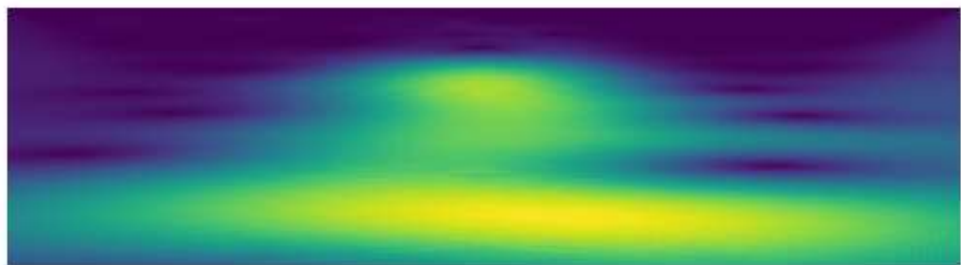
A_Scalogram:



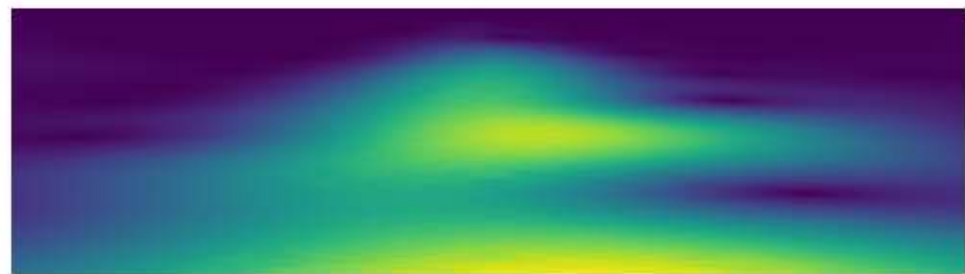
L_Scalogram



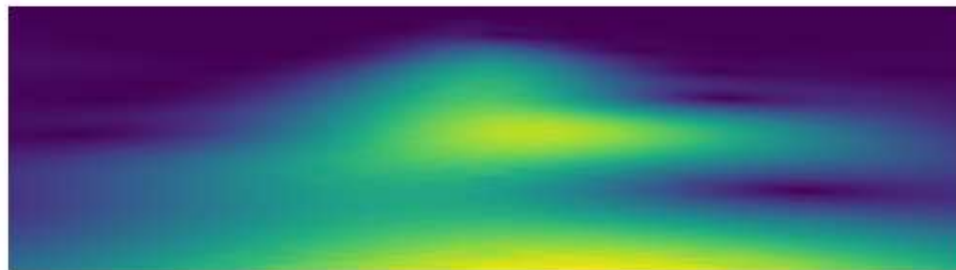
N_Scalogram



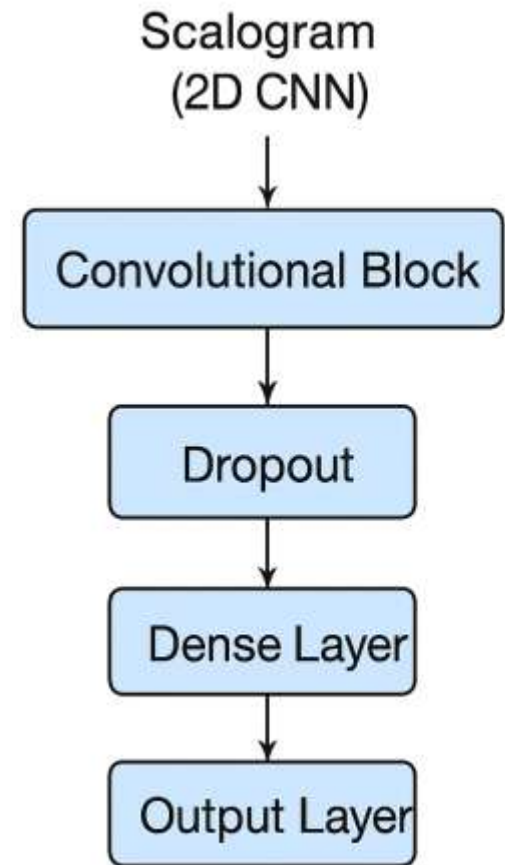
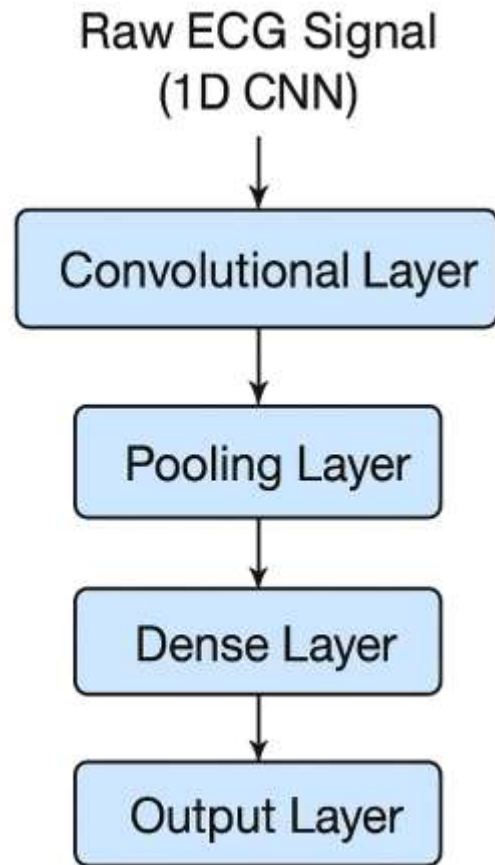
V_Scalogram



R_Scalogram:

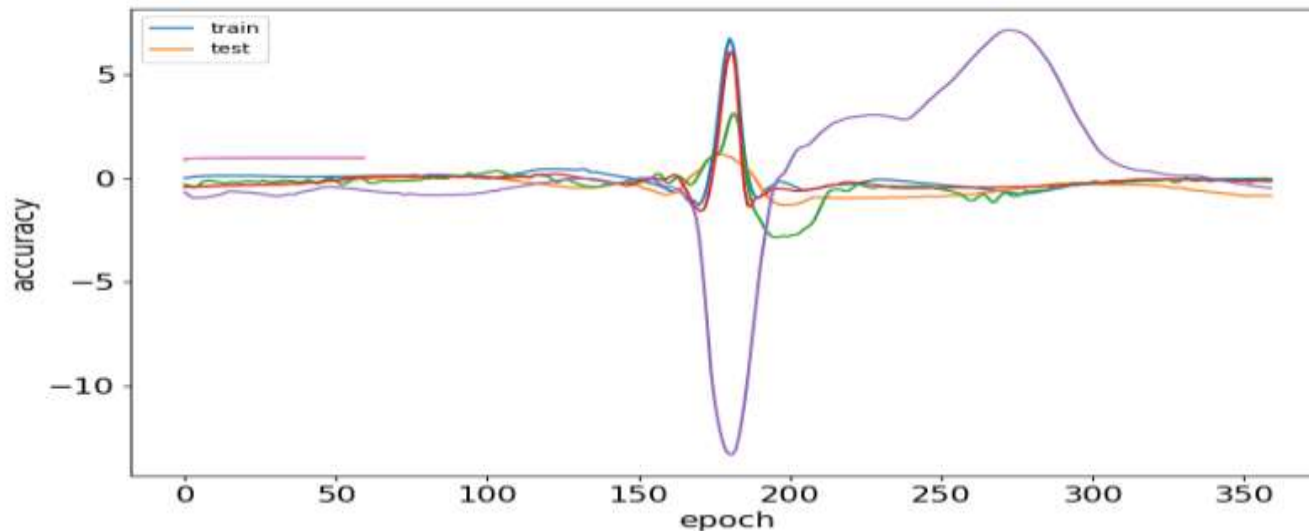


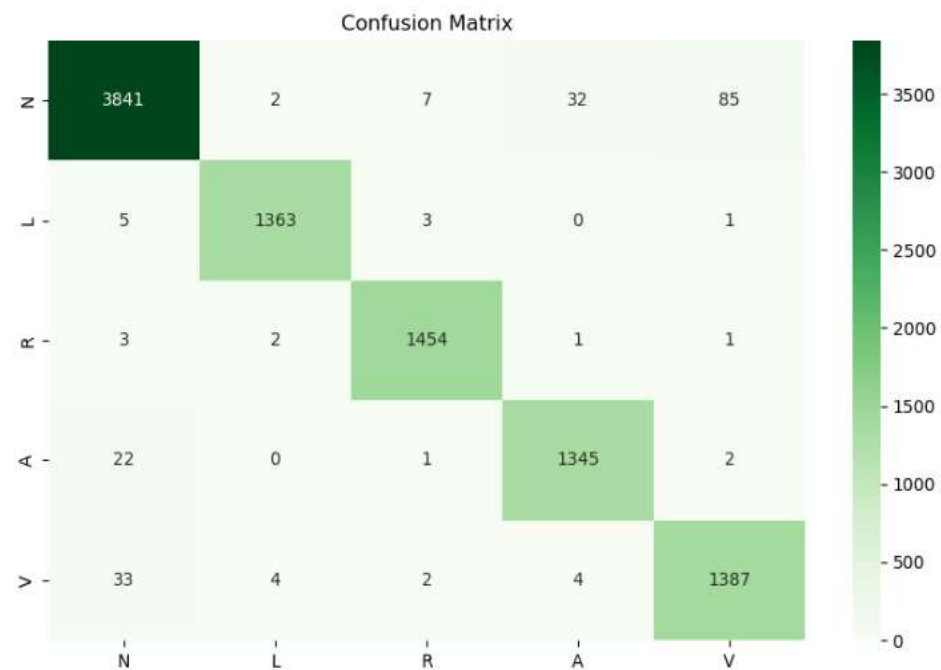
Model Architecture:



Training & Evaluation:

- Optimizer: **Adam**, Loss: **Categorical Cross-Entropy**
 - Regularization: Early stopping, dropout, learning rate scheduling
 - Metrics: Accuracy, Precision, Recall, F1-score,
-





Test Loss: 0.12198993563652039

Test accuracy: 0.9781249761581421

300/300 1s 4ms/step

	precision	recall	f1-score	support
N	0.9839	0.9682	0.9760	3967
L	0.9942	0.9934	0.9938	1372
R	0.9911	0.9952	0.9932	1461
A	0.9732	0.9818	0.9775	1370
V	0.9397	0.9699	0.9546	1430
accuracy			0.9781	9600
macro avg	0.9764	0.9817	0.9790	9600
weighted avg	0.9783	0.9781	0.9782	9600

Model: "sequential"

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 360, 16)	192
max_pooling1d (MaxPooling1D)	(None, 180, 16)	0
conv1d_1 (Conv1D)	(None, 180, 32)	6,688
max_pooling1d_1 (MaxPooling1D)	(None, 90, 32)	0
conv1d_2 (Conv1D)	(None, 90, 64)	30,784
max_pooling1d_2 (MaxPooling1D)	(None, 45, 64)	0
conv1d_3 (Conv1D)	(None, 45, 128)	139,392
max_pooling1d_3 (MaxPooling1D)	(None, 22, 128)	0
flatten (Flatten)	(None, 2816)	0
dropout (Dropout)	(None, 2816)	0
dense (Dense)	(None, 100)	281,700
dense_1 (Dense)	(None, 50)	5,050
dense_2 (Dense)	(None, 5)	255

Total params: 464,061 (1.77 MB)

Trainable params: 464,061 (1.77 MB)

Non-trainable params: 0 (0.00 B)

Results

- **1D CNN:** Reasonable baseline, but confused **N vs R**
- **2D CNN:** For better robustness, Trying to improve minority-class (L, A, V) detection
- **Balanced dataset** → reduced bias towards Normal class
- **Key Insight:** Scalogram-based 2D CNN + balanced training = best performance

PTB-XL Database

PREPROCESSING

Dataset: PTB-XL (12-lead, 10s recordings, 21k samples)

- **Challenges:** Noise & artifacts despite clinical-grade collection

- **Preprocessing pipeline:**

- Baseline wander removal → **high-pass Butterworth filter** (<0.5 Hz drift)

- Powerline noise removal → **50/60 Hz notch filter**

- Signal normalization → zero mean, unit variance per lead

- Segmentation → beat-level (R-peak) & fixed 1–2s windows

Classical ML Baselines (Feature Engineering):

- **Feature Extraction**

- Wavelet Transform (DWT) → time-frequency components
- Entropy measures (Shannon, spectral, sample) → signal complexity
- Statistical descriptors → mean, variance, skewness, kurtosis
- HRV metrics (SDNN, RMSSD, pNN50) → rhythm irregularities

- **Models Tested**

- Logistic Regression (One-vs-Rest) → simple, interpretable
- Random Forest → non-linear relations, feature importance
- LightGBM → handled imbalance better, careful tuning

Deep Learning (1D ResNet Approach):

Motivation

- End-to-end learning → no hand-crafted features
- Residual connections → deeper network, stable training
- Captures **multi-lead temporal dependencies**

Architecture

- Input: **12-lead ECG, 5000 samples (10s × 500 Hz)**
- Residual blocks (Conv + BN + ReLU + skip connections)
- Global Average Pooling → reduces overfitting
- Output: Multi-label arrhythmia classification (sigmoid layer)

Training Strategy

- Loss: **Binary Cross-Entropy**
- Augmentation: scaling, shifting, noise injection
- Cross-validation: **5-fold stratified**
- Optimizer: Adam + Cyclical LR scheduling
- Threshold tuning → maximize per-class F1

Results:

- **1D ResNet > Classical ML**

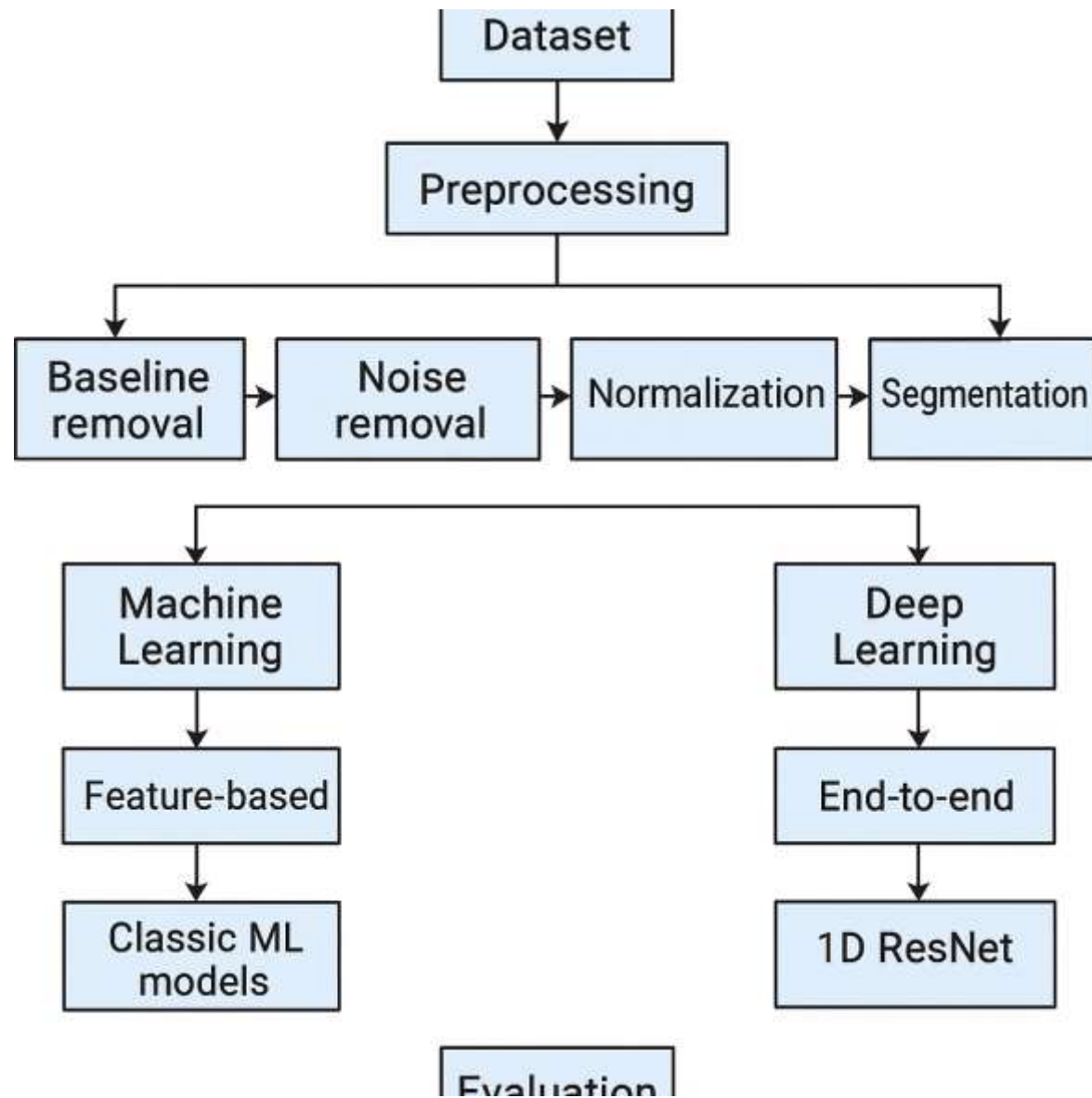
- F1-score $\uparrow$, especially for rare arrhythmias
- **AUC > 0.95** across classes
- Learned subtle multi-lead interactions

Observed Overfitting

- Training loss $\downarrow$, validation loss plateaued
- Higher training F1 than validation (esp. rare classes)

Mitigation (Ongoing Work)

- Regularization: L2 weight decay, dropout
- Advanced augmentation: lead masking, temporal warping, GANs
- Class-balanced loss: focal loss, weighted BCE
- Architecture tuning: deeper/dilated conv, attention modules



Learnings:

- ECG classification requires both **signal-domain (1D)** and **time–frequency (2D)** analysis.
- **Preprocessing** (denoising, baseline correction) is essential for reliable feature extraction.
- **Data balancing** (oversampling/augmentation) improves generalization across imbalanced classes.
- **Scalogram-based CNNs** capture richer temporal–spectral patterns than raw 1D alone.
- Benchmark datasets (**MIT-BIH** & **PTB-XL**) provide complementary strengths: small precise annotations vs. large diverse recordings.

References:

- Goldberger AL, et al. *PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals*. Circulation, 2000.
- Moody GB, Mark RG. *The MIT-BIH Arrhythmia Database on CD-ROM and software for use with it*. Computers in Cardiology, 1988.
- Wagner P, et al. *PTB-XL: A large publicly available electrocardiography dataset*. Sci Data, 2020.
- Acharya UR, et al. *A deep convolutional neural network model to classify heartbeats*. Computers in Biology and Medicine, 2017.

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THANK YOU